

Program : Diploma in Textile Technology	
Course Code : 5062	Course Title: Fashion Design & Apparel Manufacture
Semester : 5	Credits: 4
Course Category: Program core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To acquire knowledge on the process involved in Apparel manufacturing.
- To Apply and creatively adapt knowledge and skills in fashion design and technology, to practice or for future study.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Sewing Thread/Yarn Manufacture		Yarn Manufacture-II	4
Fabric Manufacture		Design and Structure of Fabrics, Fabric Manufacture -I, II	3 & 4
Garment lab		NSQF aligned workshop	3 & 4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Classify various garments and delineate patterning and grading of various garments	14	Applying
CO2	Outline the process involved in Apparel manufacturing and Assess seams, stitches and tools required for clothing construction.	15	Applying
CO3	Analyse fusing, pressing, packing, fabric inspection and garment defects.	14	Understanding
CO4	Outline on Fashion Designing.	15	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2				3			
CO3		3					
CO4			3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Classify various garments and delineate patterning and grading of various garments		
M1.01	Outline the structure of apparel industry and relate the selection of fabrics for various garments	5	Understanding
M1.02	Express the theory of human anatomy and Interpret the importance and taking body measurements	5	Applying
M1.03	Explain the importance of pattern drafting and grading	4	Applying
Contents: Structure of the Apparel Industry - Classify garments for men, women and children - selection of fabrics for various garments based on properties. Theory of Human Anatomy and its importance in apparel manufacturing. - 8 Head and 10 Head theory - Measurements and its importance - Body measurements for measuring, ladies, children's and men's garments. Pattern making - methods (Drafting & Draping) - Importance of paper patterns - types of paper patterns - Importance and principles of pattern drafting - pattern grading - Grading of various components of garments - Study of marker planning. Garment Construction Tools & Equipment - Measuring, Marking, Cutting, Pressing and General tools.			
CO2	Outline the process involved in Apparel manufacturing and Assess seams, stitches and the use of components and trims		
M2.01	Know the importance of spreading and pattern layout in apparel manufacture.	5	Applying
M2.02	Identify the cutting machines used in apparel manufacture	5	Understanding

M2.03	Discuss the sewing machines used and restate seams, stitches and trims.	5	Understanding
	Series Test - I	1	
Contents: SPREADING: Objects - methods for spreading - various types of woven fabrics - types of lay - pattern lay out - Rules in pattern layout - methods for layout - layout for asymmetrical designs - Spreading Machines - Manual its problems, semi - automatic and fully automatic -Economy of fabric in placing pattern - Rules for placement of pattern if the fabrics insufficient. CUTTING: Importance of fabric cutting - Methods of cutting - mechanism of straight knife, rotary, band knife and die cutters. Computer controlled cutting knives. SEWING: Classification of sewing machines, Brief study of Single Needle Lock Stitch sewing machine- Seams- seam types; Stitch - stitch types Selection of needles and threads for different types of garments. The use of components & trims - Labels, lining, Interlining.			
CO3	Analyse fusing, pressing, packing, fabric inspection and garment defects.		
M3.01	Gain knowledge on the process of fusing	4	Understanding
M3.02	Familiarise the process of pressing apparels	3	Understanding
M3.03	Identify materials used and methods of packing readymade garments	3	Understanding
M3.04	Summarise the defects and inspection methods used in apparel manufacture	4	Understanding
Contents: FUSING: Requirements and Process- Base cloth and resin coating system - Effect of temperature time and pressure in fusing PRESSING: the purpose of pressing and types of pressing, Pressing machineries - iron pressing, block or die pressing, buck pressing, form pressing, steam air finisher, steam tunnel - controls in pressing. PACKING: materials used for packing - Classification of Packing - Stand up pack - Flat pack - Hanger pack - Dead man pack - its importance INSPECTION: Raw material inspection - various fabric inspection system-4 point system, 10 point system - fabric defects in woven & knitted fabric, Garment defects, Final Inspection - Inspection Acceptable Quality Level (AQL) systems.			
CO4	Precise Fashion Designing.		
M4.01	Rephrase the term fashion	5	Understanding
M4.02	Extend the requirements and principles of design	5	Understanding

M4.03	Summarise e-marketing and importance of CAD/CAM	5	Understanding
	Series Test – II	1	
Contents: FASHION: Terms related to fashion - fashion, style, fad, & classic. Fashion life cycle, fashion garments vs basic garments. DESIGN: Definition, Types-structural design and decorative design - application of structural and decorative design in dresses; Elements of design - line shape, colour, size & Texture. Principles of Design in Apparels - Balance - formal and informal, Rhythm - through repetition, radiation and gradation, emphasis, harmony & proportion. Fashion market and marketing environment - introduction to fashion e-commerce, apparel and fashion e-business. Importance of CAD/CAM in the field of designing in apparel.			

Text /Reference:

T/R	BookTitle/Author
T1	Taulkdar -Garment Technology - Wood head publishing Ltd
R2	Garment Design NIFT/New Delhi
R3	Garment Manufacture NIFT/New Delhi
R4	Chuta A J Introduction to Clothing Production Management
R5	Cookin G Introduction to Clothing Manufacture
R6	Carr.H The technology of Clothing manufacture

Online resources:

Sl.No	Website Link
1	http://nptel.ac.in/
2	https://www.youtube.com/
3	www.textilelearner.blogspot.com
4	https://swayam.gov.in/
5	http://eacharya.inflibnet.ac.in/vidya-mitra/